EigenX-v2 prescribed path: scoped obstruction and conditional control

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Batch: `BATCH-be2a37`
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Result at a glance

[Independently reviewed] `EigenX-v2-Path-FO-KEM` is not admitted as a complete-EndRing-secure PKE or KEM. All five independently owned joints break; the composed obligation vector is `O1 open, O2 open, O3 open, O4 contradicted, O5 contradicted, O6 open`. There are zero reviewed discharges. The construction and material-difference gates fail [S10--S15].

[Independently reviewed] The decisive mechanism result is narrower than a universal impossibility theorem. For this exact Seta-style `M=1`, endpoint-plus-full-`N`-torsion interface, every accepted decapsulation factors through

```
q_N = (d,e,theta(P_N),theta(Q_N)),
Decaps((pk,W,z),c) = Decaps_Q((pk,q_N,z),c).
```

The unused remainder of the saturated evaluable rank-four witness `W` can be erased after `q_N` is computed. This proves that the complete ring is not necessary for the declared recipient operation; it does not prove that `q_N` cannot be lifted to `W`, and it says nothing universal about differently typed EndRing mechanisms [S5, S10, S14].

[Independently reviewed] The required security-reduction direction is also absent. A complete EndRing solver may enable decryption, but that is the reverse attack route. No reduction embeds a bare complete-EndRing challenge into the exact real key law, simulates decapsulation without `W`, or converts one attacker bit or one recovered path into the arbitrary-input OneEnd/Isogeny oracle needed to reconstruct a complete ring [S6, S11, S14].

[Conditional control only] `CTRL-CEST-AUTH-XFER` is a separate accepted-key, single-user, classical-ROM control under `CEST-XFER-COMP`. It conditionally closes the old raw-XOR bit-flip when paired with the stated one-time IND-CCA DEM, but it factors through `EvalLine_(P,A)`, is not EigenX-v2, is not EndRing-bound, and cannot repair `O4`, `O5`, or the material-difference gate [S7, S12, S14].

[Unmeasured] This was a zero-run symbolic and source audit: no scientific, implementation, experiment, formalizer, benchmark, or parameter-selection run was performed. No parameter set, quantitative performance graph, practicality, deployment, or post-quantum claim is present [S14]. The mechanism and reduction boundaries are shown in editable Mermaid and browser-rendered SVG.

Evidence labels and scope

- **[Proposed]** identifies the frozen producer candidate, interface, or theorem target.
- **[Derived]** records a symbolic consequence of frozen definitions, not a measurement.
- **[Independently reviewed]** is owned by a valid J1--J5 review and copied without voting by the final composition.
- **[Source-supported conditional]** is a literature result only under its exact hypotheses.
- **[Conditional control only]** belongs to the separate CEST control, not to EigenX-v2.
- **[Unmeasured]** marks the absence of experimental or performance evidence.

This report follows the frozen methodology. It assesses one prescribed-path/torsion-binding candidate and one separately named control. It does not construct a new scheme or quantify over all EndRing PKE/KEM designs.

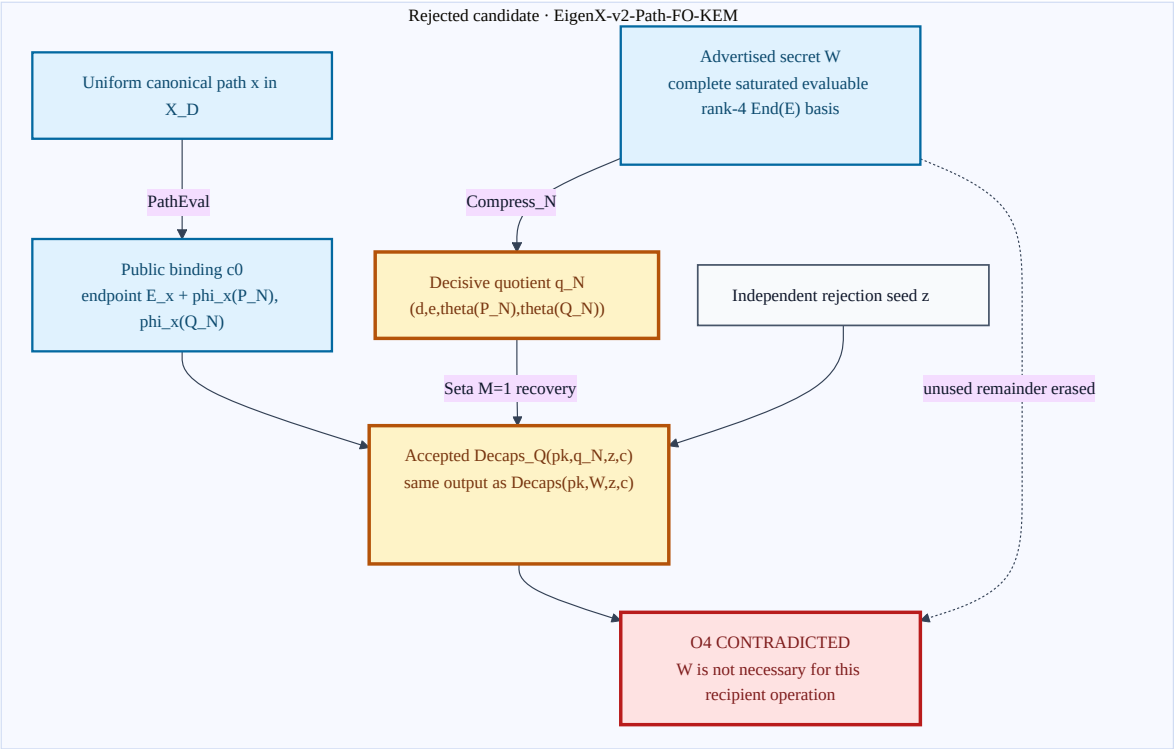


Figure 1a. Candidate mechanism and decisive q_N factorization.



Figure 1b. Required reduction: reverse route and exact real-key embedding failure.

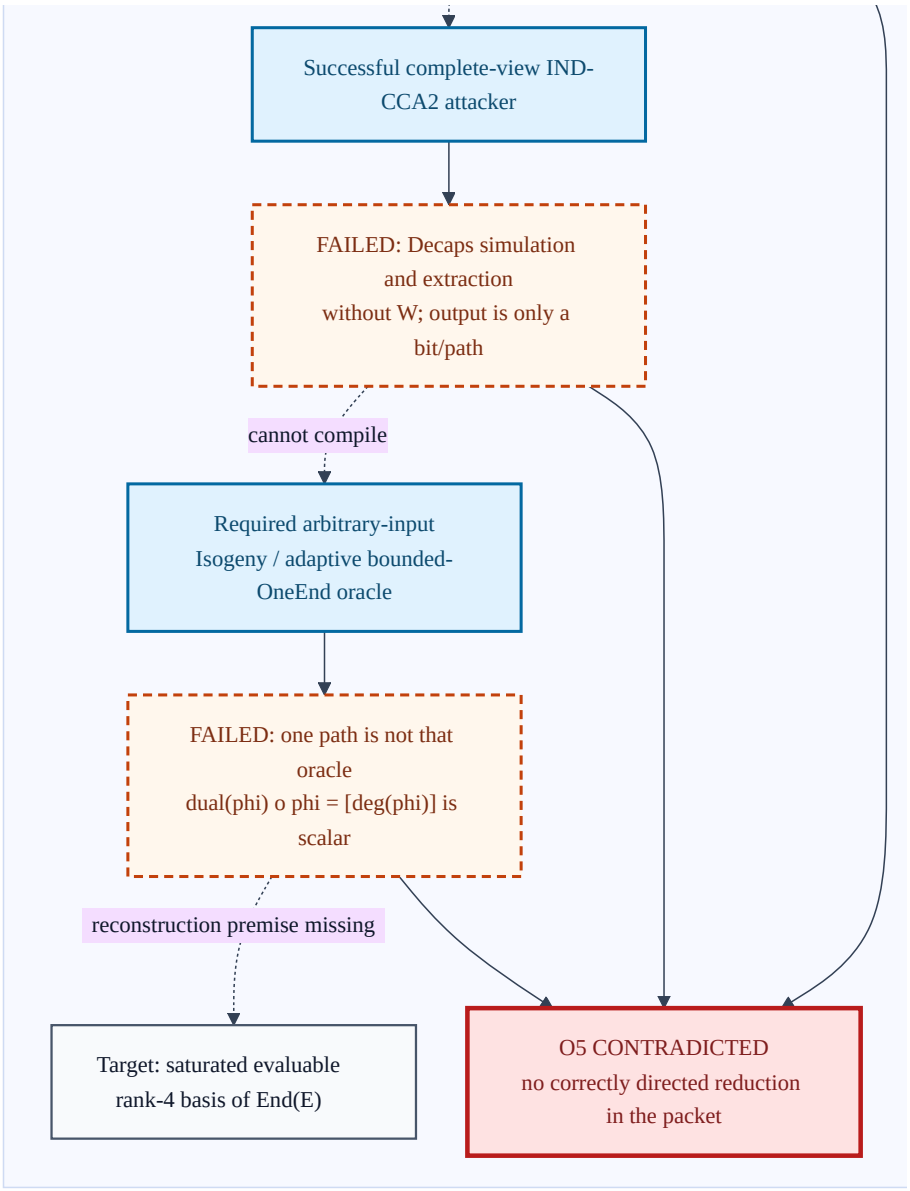


Figure 1c. Required reduction: attacker simulation, oracle-compiler failure, and O5 contradiction.

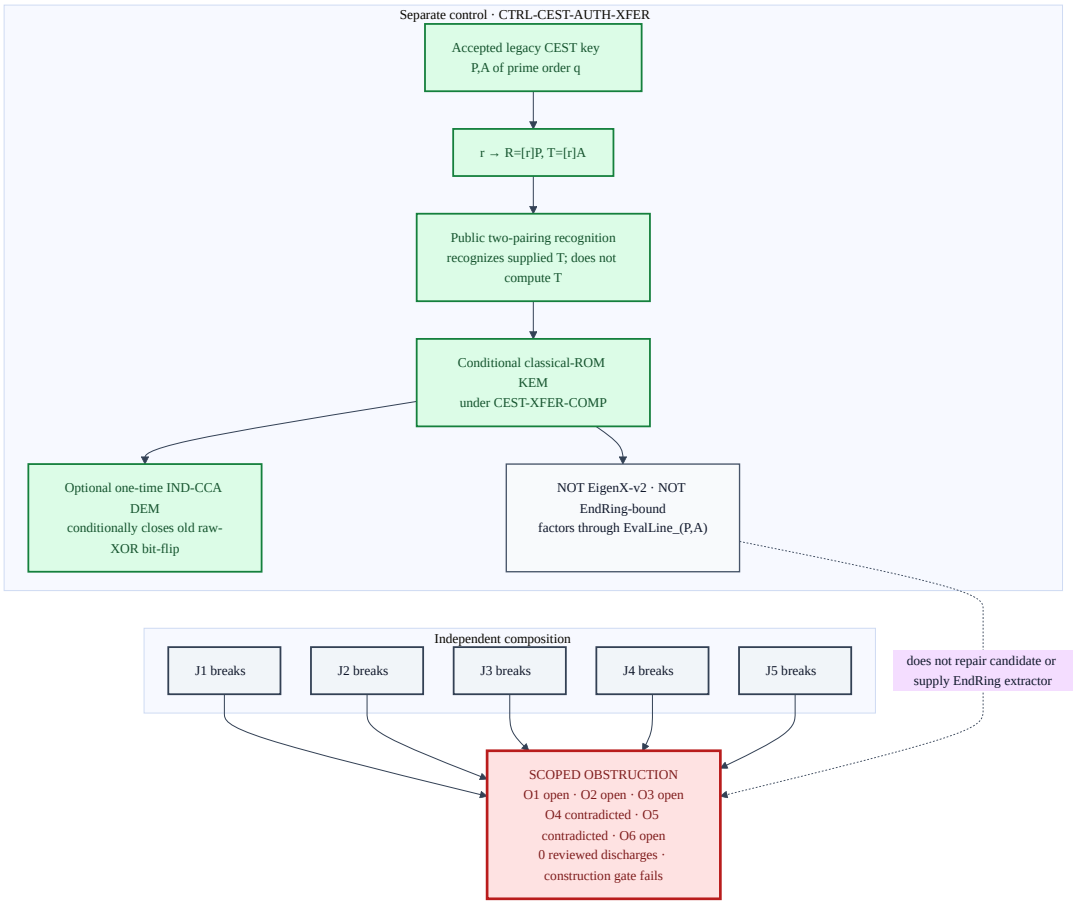


Figure 1d. Separate conditional CEST control and the five-joint composed outcome.

1. Frozen candidate and separate control

Rejected candidate: `EigenX-v2-Path-FO-KEM`

[Proposed] The internal deterministic `PathPKE` samples a canonical nonbacktracking degree-`D` path `x`, evaluates `phi_x:E -> E_x`, and binds the path with the endpoint and the images of a full public `N`-torsion basis:

```
c0 = Encode(version, Hash(pk), CanCurve(E_x),
            phi_x(P_N), phi_x(Q_N)).
```

Under the Seta conditions, including `N^2 > 4D`, endpoint plus full torsion action is injective for this family and Seta Algorithm 1 conditionally recovers the path/kernel. Endpoint-only injectivity or recovery is not supplied. The HHK transformation `U` then conditionally gives a classical-ROM IND-CCA KEM from a `delta_1`-correct OW-PCA base PKE over uniform `X_D` [S4, S5, S13].

The intended secret is `(pk,W,z)`, where `W` is a saturated rank-four basis of exactly `End(E)` with efficient evaluators, and `z` is the independent uniform implicit-rejection seed. This is a conditional interface, not an executable scheme: `TrapCurveGen_ER`, `SelectTrap`, canonical endpoint/point transport, validation, total `Dec0`, the exact real key law, and bounded retry/cost accounting remain unavailable [S4, S5, S12].

Separate control: `CTRL-CEST-AUTH-XFER`

[Conditional control only] The control uses an accepted legacy CEST key, samples `r`, publishes `R=[r]P`, derives `T=[r]A`, and hashes canonical `(pk,R,T)` in a classical random oracle. Public pairings uniquely recognize a *supplied* `T`:

```
e_q(P,T)=e_q(R,A)  and  e_q(A,T)=1
    iff T=[r]A,
```

given the stated prime-order membership and nondegeneracy conditions. The recognizer does not compute `T`. The frozen producer reduction conditionally establishes

```
Adv_KEM(A) <= Succ_CEST-XFER-COMP(B)/2,
Adv_PKE(A) <= Succ_CEST-XFER-COMP(B_X) + Adv_otIND-CCA_DEM(B_D).
```

This control is useful only at its named XFER/EvalLine boundary. Its legacy key-generation law remains incomplete, and no transfer-solver-to-complete-ring reduction is supplied [S7, S12, S14].

2. Decisive functionality factorization

[Independently reviewed] Let

```
Compress_N(W) = q_N = (d,e,theta(P_N),theta(Q_N)).
```

For the frozen `M=1` recovery mechanism, `Dec0` consumes only `q_N`, public data, and the ciphertext torsion images. The complete accepted KEM behavior therefore factors through `(pk,q_N,z)`. Two complete witnesses with the same quotient, or one full witness and the quotient alone, produce the same accepted decapsulation outputs [S10, S14].

This is a functionality quotient, not a public inverter. Publishing `q_N` would also publish the recipient recovery operation and destroy the intended asymmetry. Keeping it secret yields the finite-action Seta/TCSSI-style trapdoor; it does not make the unused part of `W` necessary. The control is strictly scoped:

Route	Reviewed disposition	Boundary
Endpoint plus full `N`-torsion images	Conditionally functional, but `W` is unnecessary	Seta injectivity/recovery uses only `q_N` under its conditions.
Endpoint only	Inconclusive	The only retrieved injectivity and polynomial inversion theorem is lost.
Public `q_N`	Negative confidentiality control	The public can perform recipient recovery.
Full `W` to decrypt	Reverse attack route	Sufficiency for attack is not a security reduction.

No reviewed artifact proves that `q_N` is easy to compute, that it cannot imply `W`, or that every EndRing-bound recipient mechanism has such a quotient.

3. Required reduction direction and exact failures

The required direction is:

```
complete-EndRing challenge E + successful complete-view IND-CCA2 attacker
    -> saturated efficiently evaluable rank-four basis of End(E).
```

[Independently reviewed] The frozen chain fails at distinct typed arrows [S6, S11, S14]:

1. A bare challenge curve does not include `W`, an unknown prescribed path, or the auxiliary state needed to sample the exact `TrapCurveGen_ER` public key law. No equal- or bounded-distance challenge embedding is supplied.
2. Without `W` or an equivalent recovery interface, the reducer cannot simulate the exact decapsulation oracle on every allowed nonchallenge byte string.
3. The adversary returns a distinguishing bit. Even granting extraction of one accepted path, a connecting path is not a non-scalar endomorphism of the challenge curve; composing it with its dual gives the scalar `[deg(phi)]`.
4. Page--Wesolowski reconstruction requires valid bounded-OneEnd answers on every curve queried adaptively. One KEM sample or one recovered path does not instantiate that arbitrary-input oracle.

The opposite direction,

```
complete EndRing solver -> derive trap action/path -> break the KEM,
```

is an attack upper bound. It cannot be reversed by relabeling. The exact frozen IND-CCA2 failure is therefore **non-instantiation**, not a demonstrated ciphertext attack against a completed EigenX-v2 implementation: HHK Theorem 4 only applies after `PathPKE` is a PPT PKE with a quantified real-law correctness bound and OW-PCA security. Those premises are not established [S11--S14].

For contrast, the old unauthenticated CEST syntax had a one-query raw-XOR bit-flip. The separate authenticated CEST control removes that exact payload shape and conditionally covers same-`R`/changed-DEM queries under its one-time IND-CCA DEM assumption. This does not transfer security to EigenX-v2 [S7, S12].

4. Exact O1--O6 composition

The final composition applies the obligations conjunctively. No joint votes against, substitutes for, or repairs another [S14].

Obligation	Composed status	Exact composed finding
O1 — exact hard problem	`open`	CompleteEndRing-KG can be typed, but the exact real KeyGen public-key law, accepted-key conditioning, public auxiliary-data distribution and a distribution-equal or bounded challenge embedding are unavailable.
O2 — honest algorithms	`open`	The shell names algorithms and failure branches, but `TrapCurveGen_ER`, `SelectTrap`, canonical endpoint/point transport, total `Dec0`, validation, exact costs and bounded retry behavior are not executable complete algorithms on the real law.
O3 — quantified correctness	`open`	Accepted-key/per-path correctness is conditional. No terminating exact KeyGen law or quantified `delta_1` bound over the actual correctness experiment is established, so the producer's proposed discharge is not promoted to a reviewed discharge.
O4 — recipient mechanism	`contradicted`	For the exact `M=1` endpoint-plus-full-`N`-torsion interface, Decaps factors through `q_N=(d,e,theta(P_N),theta(Q_N))` and `z`. The complete saturated evaluable rank-four `End(E)` witness is not necessary for the declared recipient operation, and the algebraic core is Seta's finite-action prescribed-path mechanism.
O5 — correctly directed reduction	`contradicted`	No correct-direction attacker-to-complete-EndRing reduction exists in the packet. The EndRing challenge does not supply an unknown prescribed path; the decapsulation oracle cannot be simulated without the withheld `W`; and a recovered single path or distinguisher bit does not instantiate the arbitrary-input OneEnd/Isogeny oracle required for complete-ring recovery. This contradicts the tested reduction for this frozen interface, not every possible new reduction or mechanism.
O6 — complete adversarial view	`open`	The packet has a substantial symbolic classical-ROM inventory and rejects free auxiliary advice, but the real-law adversarial distribution, canonical malformed-input grammar, failure schedule, exact bit/operation/memory/preprocessing costs, source-complete comparison and implementation observables remain incomplete.

The vector has `0` reviewed discharges, `4` open obligations, and `2` contradicted obligations. The candidate disposition is `scoped_obstruction`; complete IND-CCA2, a correctly directed EndRing reduction, novelty/material difference, and implementation eligibility are not established [S14].

5. Positive, negative, and inconclusive routes

Positive only within stated conditions

- **[Source-supported conditional]** Seta supports endpoint-plus-full-torsion injectivity and recovery for its exact family and conditions, and explicitly identifies the `M=1` finite torsion-action requirement [S10, S13].
- **[Source-supported conditional]** HHK Theorem 4 supports the displayed OW-PCA-to-IND-CCA outer transform in the classical ROM, with `Adv_IND-CCA <= q_H/|X_D| + Adv_OW-PCA` and correctness error `delta_1`, if its base-PKE premises hold [S12, S13].
- **[Conditional control only]** The CEST recognition-table simulation is a coherent accepted-key classical-ROM route to the separately named `CEST-XFER-COMP` assumption [S7, S12].

Negative for this exact frozen interface

- Complete `W` is not necessary for accepted decapsulation: `q_N` suffices.
- The algebraic core is not materially different from Seta's finite-action prescribed-path mechanism.
- No correctly directed scheme-attacker-to-complete-EndRing reduction is supplied.
- The construction gate and material-difference gate fail [S10--S14].

Inconclusive and still open

- Endpoint-only exact-path injectivity and polynomial full-ring recovery.
- A distribution-matched `TrapCurveGen_ER` and semantic `SelectTrap` with total polynomial algorithms, retry tails, and complete evaluator costs.
- Any new correctly directed reduction or materially different EndRing recipient mechanism.
- Whether a separately proved finite-action-to-complete-ring lift exists on the exact real key law.

Missing evidence is not evidence of universal nonexistence [S14, S15].

6. Exact independent owner verdicts

The following verdict text is copied exactly from the sole joint owners through the final composition, without voting [S14].

J1 — `breaks`

> The recipient operation is conditionally well typed as a Seta-style prescribed-path inverter, and the public key alone does not provide that inverter. The joint nevertheless breaks because the complete evaluable rank-four witness W is not necessary for the declared mathematical Decaps operation. For every accepted key, all W-dependent recovery data can be cached as q_N=(d,e,theta(P_N),theta(Q_N)); complete KEM behavior then factors through (pk,q_N,z), where z is the independent implicit- rejection seed. This is a scoped result for this exact M=1 Seta-style endpoint-plus-torsion interface, not a universal result about EndRing PKE.

J2 — `breaks`

> J2 only. The allowed opening does not supply a typed reduction with the required direction. This is not a whole-candidate verdict and is not an impossibility result for every prescribed-path or EndRing-based KEM.

J3 — `breaks`

> The frozen EigenX-v2-Path-FO-KEM packet does not instantiate a complete IND-CCA KEM. HHK Theorem 4 supports the displayed OW-PCA-to-IND-CCA outer transform conditionally, but the base PathPKE is not a PPT PKE with a quantified correctness law: TrapCurveGen_ER, SelectTrap, canonical endpoint/point transport, validation, and total Dec0 recovery remain unavailable. The malformed and related-ciphertext rules are therefore a conditional interface rather than an executable complete syntax. This is a verdict only on J3, not on the other review joints or the whole claim. The separate authenticated CEST packet survives J3 as a coherent accepted-key, classical-ROM conditional control under CEST-XFER-COMP; it does not repair EigenX-v2 or supply an EndRing theorem.

J4 — `breaks`

> On the frozen zero-run packet, the producer has a substantially complete symbolic checklist for a single-user classical-ROM cost/view audit and correctly rejects zero-cost or view-free auxiliary advice. It establishes no instantiated total cost or complete real-key adversarial view. All concrete, asymptotic, implementation and comparative cost claims remain open, and the conclusion is limited to this prescribed-path/torsion-binding interface.

J5 — `breaks`

> J5 breaks any admission of EigenX-v2-Path-FO-KEM as a materially new, complete-EndRing-secure construction. This is a scoped source-overlap and assumption-separation verdict, not a universal impossibility theorem for all EndRing PKE or KEM designs.

7. Source normalization and corrected metadata

[Independently reviewed] Seta dominates the functional algebraic core. SiGamal/C-SiGamal and the inspected key-updatable PKE work are adjacent but use different operations and assumptions. The retrieved LIT-SiGamal abstract is insufficient for a theorem-level comparison. Page--Wesolowski supplies relevant problem reductions but not exact sender-path recovery or the missing IND-CCA-attacker extractor. HHK supplies only the conditional outer transform [S13, S14].

The valid J5 review corrected three producer titles and normalized a fourth citation [S13, S14]:

1. Aurel Page and Benjamin Wesolowski, “The Supersingular Endomorphism Ring and One Endomorphism Problems are Equivalent,” EUROCRYPT 2024, LNCS 14656, pp. 388--417, DOI `10.1007/978-3-031-58751-1_14`.
2. Edward Eaton, David Jao, Chelsea Komlo, and Youcef Mokrani, “Towards Post-Quantum Key-Updatable Public-Key Encryption via Supersingular Isogenies,” SAC 2021, LNCS 13203, pp. 461--482, published 2022, DOI `10.1007/978-3-030-99277-4_22`.
3. Pratima Jana and Ratna Dutta, “UPKE and UKEM Schemes from Supersingular Isogenies,” IACR ePrint 2025/1010, 2025.
4. Tomoki Moriya and Miha Stopar, “LIT-SiGamal: An efficient isogeny-based PKE based on a LIT diagram,” IACR ePrint 2024/521, 2024; the landing page calls it a minor revision of the ASIACRYPT 2025 publication.

LIT-SiGamal full text was unavailable: the ePrint PDF returned HTTP 403 and the Springer route returned HTML rather than an inspectable PDF. The ePrint 2025/1010 PDF also returned HTTP 403. Landing-page metadata establishes none of novelty, non-overlap, equivalence, or EndRing security.

8. Review independence and J5 replacement

The valid five-report independence checker returned:

```
PASS: 5 report(s), every joint owned and attested, blindness respected, controls declared
```

This is a mechanical ownership/attestation check, not a scientific verdict [S14]. J5 attempt 1, owned by `eigenx-v2-j5`, had a strict mutual-blindness deviation: a filename-format search mechanically scanned current J1/J3 file names without displaying their contents. The attempt was abandoned before any artifact was created and has no scientific ownership. J5 attempt 2, owned by `reviewer-eigenx-j5-clean`, was an independent replacement, completed cleanly, accepted by the checker, and is the sole scientific J5 owner [S14, S16, S17].

9. Zero-run accounting, limitations, and canonical problem

Activity	Count
Scientific runs	0
Implementation runs	0
Experiment runs	0
Formalizer runs	0
Benchmarks	0
Parameter-selection runs	0
Measured observations	0

Reading, hashing, YAML/JSON parsing, running the review-independence checker, composing owner verdicts, diagram/PDF rendering, and visual inspection are administrative documentation operations rather than scientific runs [S14]. Because the batch contains no measurements, a quantitative chart would imply evidence that does not exist; Figure 1 is an explanatory typed-arrow diagram.

The canonical problem remains EigenX/CEST EndRing reduction is not established, with status `investigating`; its exit criteria are not met [S15]. Exact limits:

- This result covers only the frozen prescribed-path/torsion interface and is not a universal impossibility result for EndRing PKE/KEM.

- Missing algorithms, source full text, parameters, costs, and lemmas remain open evidence obligations, not proof that replacements cannot exist.
- Conditional accepted-key correctness and an applicable outer transform do not establish complete syntax, unconditional correctness, or EndRing security.
- The authenticated CEST result is a separate XFER/EvalLine control in the single-user accepted-key classical ROM; it is neither the candidate nor a discharge of the batch objective.
- No parameter set, bit-security estimate, implementation, benchmark, practicality, deployment, QROM, multi-user, or post-quantum security claim is established.

10. Exactly one nonautomatic action

`TASK-20261013-26410a` — produce this frozen Markdown, editable Mermaid, browser-rendered SVG, inspected PDF, and visual-inspection record without adding a scientific result or security label. This report set completes that one nonautomatic action. After the already queued J6 documentation review and ledger archive, the frozen stop boundary is to close `BATCH-be2a37`; no successor construction round is authorized without a new explicit coordinator decision and user-authorized scope [S14].

Sources and inspected scope

Ref	Source	Provenance and scope used here
S1	Opening contract	`internal`; full objective, candidate/control boundary, obligations, and zero-run rule.
S2	Review plan	`internal`; J1--J5 ownership, mutual blindness, controls, and composition rule.
S3	Producer snapshot	`internal`; content-first frozen paths and producer hashes.
S4	Candidate packet	`internal`, producer; full candidate syntax, factorization argument, controls, and limits.
S5	Scheme contract	`internal`, producer; typed O1--O6 contract and conditional HHK statement.
S6	Reduction boundary	`internal`, producer; exact arrow types, earliest gaps, source-interface mismatch, and controls.
S7	Authenticated CEST control	`internal`, producer; accepted-key classical-ROM XFER control and optional DEM composition.
S8	Cost and adversarial-view packet	`internal`, producer; complete symbolic checklist with uninstantiated values.
S9	Source-overlap packet	`internal`, producer; Seta, SiGamal, UPKE, LIT-SiGamal, PW24, and HHK comparison with retrieval limits.
S10	J1 review	`internal`, independently reviewed; complete-ring necessity and factorization.
S11	J2 review	`internal`, independently reviewed; reduction direction, embedding, simulation, and extraction.
S12	J3 review	`internal`, independently reviewed; complete syntax, correctness, HHK scope, and CEST control.
S13	J4 review	`internal`, independently reviewed; cost and adversarial-view audit.
S14	J5 review and final composition	`internal`; source normalization plus exact non-voted joint/obligation composition and limits.
S15	Canonical problem	`internal`; unresolved named reduction gap and exit criteria.
S16	J5 attempt-1 release	`internal`; abandoned administrative deviation, no artifact.
S17	J5 attempt-2 release	`internal`; completed clean replacement and bound valid J5 artifact.